Element G: Construction of a Testable Prototype

Our school's students that attend sports after school that participate in strength and conditioning; this is where students lift and do indoor exercises, need more time, space, and equipment in the weight room. This is because there is not enough time given only 45 minutes to productively work out in the gym, with the limited amount of time given. The problem is that all of our current solutions either take a faculty member's time or more money to create. The people involved include Mr. B, who is the athletic director, our school's students, these are the students who use the gym and our school as a whole. The issue is occurring during afternoon programming, which is an issue due to a lack of time for students to work out.

Our idea is unique in the way that we aren't building a physical prototype to solve our problem. For our "prototype," we made a list of materials needed, including new weights and a worked-out schedule of when and where students will be and what they will be doing. In our schedule, we needed to make sure that it fit the teachers/coaches' time restraints. At first, we had the idea of extending the time in the schedule, but this was quickly changed due to the knowledge that teachers will not be able to stay longer at the weight room. We then decided that it was unnecessary to have doubles of all weights, only doubles of the most used and common ones, which will help save the school money.

We decided to go with a new schedule in order to hopefully help with the cramming in the weight room. We created a new schedule and researched new equipment; other than that, we didn't use any other materials. In our schedule, we show that by separating the groups out into smaller numbers, it would benefit the efficiency in the weight room. We asked multiple people about their opinions on this schedule, using a survey-style format. We used this information to help us learn from others and hopefully help them.

Below is our final schedule. We decided to change it to being divided by number instead of grade. By being divided by a random number of different students, this will give the younger students the opportunity to learn from those with more experience. If we were to divide by grade instead of numbers, there would be many more students in one group than another, which would be less productive.

Group A	Group B	Group C
3:45-4:15 → weights 4:15-4:45 → outside weights 4:45-5:15 → cardio exercise	3:45-4:15 → outside weights 4:15-4:45 → weights 4:45-5:15 → cardio exercise	3:45-4:15 → cardio exercise 4:15-4:45 → outside weights 4:45-5:15 → weights

When interviewing students who go to our school, some liked the idea of adding more hand weights, but most suggested adding a stairmaster, another lat pulldown machine, and a leg extension machine

It is important for us to understand the students' perspective on this idea, just as it is important for any engineer to understand the full aspect of their surroundings before creating change. If people don't want change, then we shouldn't change it.

We surveyed some students to help get an understanding of how they liked our ideas. We will test how the students feel about the potential schedule and new equipment. We hope that it will allow students to have more time to be actively participating in exercise. We will not test the time that coaches will work or the safety of this plan, but it will be worked into our plan.